Sovereign Default Risk in the Eurozone

A Multidisciplinary Assessment of
PIIGS versus Core EU Nations, 2025–2026

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Abstract

We conduct a multidisciplinary analysis of sovereign default risk in the Eurozone, contrasting the historically peripheral *PIIGS* economies (Portugal, Ireland, Italy, Greece, Spain) with the traditional *core* (Germany, France, Netherlands, Austria, Belgium). Drawing on debt-sustainability theory, credit-spread models, political-economy frameworks, game-theoretic bailout mechanics, and statistical risk scoring, we establish three principal findings. First, the classical PIIGS/core binary has largely dissolved: Portugal, Greece, and Ireland have achieved fiscal positions that rank among the strongest in the Eurozone, while France and Belgium exhibit debt trajectories that in earlier decades would have attracted crisis-era terminology. Second, Italy remains the single systemic risk of first order, combining the largest absolute stock of sovereign debt in the Eurozone with structural stagnation and political fragility. Third, the emergence of the *BIFS* acronym (Britain, Italy, France, Spain) in April 2026 reflects a convergence of spread-widening pressures driven by defence expenditure commitments, residual inflation shocks, and political minority-government dynamics. We formalise a Composite Sovereign Risk Index (CSRI), derive a debt sustainability condition under stochastic growth, and present a maximum likelihood estimator for default probability, supported by TikZ diagrams, `booktabs` tables, and an algorithmic risk-scoring procedure.

The paper ends with “The End”

Keywords: sovereign default, PIIGS, Eurozone, credit spreads, debt sustainability, political risk.

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1 Introduction

The acronym PIIGS — Portugal, Ireland, Italy, Greece, Spain — entered financial discourse during the European sovereign debt crisis of 2010–2012, denoting a cluster of Eurozone members whose borrowing costs soared, whose credit ratings were serially downgraded, and for whom contagion effects were most acute (De Grauwe, 2013; Reinhart and Rogoff, 2009). At the height of the crisis, ten-year yield spreads over the German Bund exceeded 500 basis points for Greece and 700 basis points at their peak (ECB, 2012). The European Central Bank's (ECB) "whatever it takes" intervention in July 2012 (Draghi, 2012) and the subsequent establishment of the European Stability Mechanism (ESM) provided conditional relief, but the structural questions — debt overhang, current account adjustment, and political economy of austerity — remained.

By 2025–2026, the landscape has been transformed in ways that challenge received wisdom.

1. Ireland and Portugal now carry some of the lowest debt-to-GDP ratios and fiscal surpluses in the Eurozone.

2. Greece, once the epicentre of the crisis, has achieved a sustained primary surplus and an investment-grade credit rating.

3. France, traditionally classified as a core economy, carries a deficit of -5.1% of GDP and a debt trajectory rising toward 129% of GDP by 2030 (IMF, 2025a).

4. Belgium's debt, at 107.5% of GDP in 2025, is projected to reach 122.6% by 2030 (Euronews, 2025).

5. A new acronym — BIFS (Britain, Italy, France, Spain) — emerged on trading desks in April 2026 to describe simultaneous spread-widening (European Business Magazine, 2026).

This paper proceeds as follows. Section 2 lays the theoretical foundations: debt sustainability, credit spread decomposition, and game-theoretic bailout mechanics. Section 3 develops the statistical and machine-learning framework for estimating default probability. Section 4 addresses the political-economy and geopolitical dimensions. Section 5 presents the empirical evidence with tables and figures. Section 7 constructs the Composite Sovereign Risk Index. Section 8 gives the risk-scoring algorithm. Section 10 concludes.

2 Theoretical Foundations

2.1 Debt Sustainability: The Intertemporal Budget Constraint

Let B_t denote the stock of government debt at time t, Y_t real GDP, r_t the real interest rate on outstanding debt, and g_t the real GDP growth rate. Define the debt ratio $d_t \equiv B_t/Y_t$. The law of motion is:

$$d_{t+1} = \frac{1+r_t}{1+g_t}\, d_t - s_t, \tag{1}$$

where $s_t \equiv (T_t - G_t^{\text{primary}})/Y_t$ is the primary surplus ratio. Debt is *sustainable* if and only if the present value of future primary surpluses equals the current debt stock:

$$d_t = \sum_{k=1}^{\infty} \prod_{j=1}^{k} \left(\frac{1+g_{t+j}}{1+r_{t+j}} \right) s_{t+k}. \tag{2}$$

Definition 2.1 (Debt-Stabilising Primary Balance)**.** The *debt-stabilising primary balance* (DSPB) is the value s^* such that $d_{t+1} = d_t$, given constant r and g:

$$s^* = \frac{r-g}{1+g}\, d_t. \tag{3}$$

Remark 2.2. When $r < g$ (the "dynamically efficient" regime does not hold), any primary balance — even a deficit — is consistent with a declining debt ratio. Italy's nominal growth rate has rarely exceeded its sovereign borrowing cost since 2015, making equation (3) binding at high levels of s^*.

2.2 Stochastic Debt Dynamics

Allow g_t to follow an AR(1) process:

$$g_t = \bar{g} + \rho(g_{t-1} - \bar{g}) + \varepsilon_t, \quad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_g^2), \tag{4}$$

with $|\rho| < 1$. Let $\phi \equiv (1+r)/(1+g)$. Under the linear approximation $(1+r)/(1+g) \approx 1 + r - g$ for small r, g, the n-period-ahead debt ratio satisfies:

$$d_{t+n} = \phi^n d_t - \sum_{k=0}^{n-1} \phi^{n-1-k} s_{t+k+1} + \sum_{k=0}^{n-1} \phi^{n-1-k} \eta_{t+k+1}, \tag{5}$$

where η_{t+k} captures the stochastic component of the interest-growth differential. The probability that d_{t+n} exceeds a critical threshold $\bar{d}$ — the "fiscal limit" — is then:

$$\Pr\big(d_{t+n} > \bar{d}\big) = \Phi\left(\frac{\phi^n d_t - \bar{d} - \mu_s^{(n)}}{\sigma_s^{(n)}} \right), \tag{6}$$

where Φ is the standard normal CDF, $\mu_s^{(n)}$ is the cumulative expected surplus path, and $\sigma_s^{(n)}$ the associated standard deviation. This expression directly underpins the risk-scoring procedure in Section 8.

2.3 Sovereign Spread Decomposition

The yield on sovereign bond i relative to the risk-free rate (Bund) decomposes as:

$$\underbrace{y_i - y_{\mathrm{DE}}}_{\text{spread}} = \underbrace{\lambda_i \cdot \mathrm{LGD}_i}_{\text{credit premium}} + \underbrace{\pi_i}_{\text{redenomination risk}} + \underbrace{\ell_i}_{\text{liquidity premium}} + \underbrace{\xi_i}_{\text{residual}}, \tag{7}$$

where λ_i is the risk-neutral default intensity, LGD_i is loss given default, π_i is the break-up premium (euro exit risk), and ℓ_i is the bid-ask-spread-based liquidity premium. Empirically, π_i was a significant component during 2010–2012 but has declined materially following the ECB's Outright Monetary Transactions (OMT) commitment and the Transmission Protection Instrument (TPI) introduced in 2022 (ECB, 2022).

Proposition 2.3 (Spread Convergence Condition)**.** *If the ECB credibly commits to unlimited OMT/TPI intervention at spread $\bar{s}$, then in a rational expectations equilibrium the equilibrium spread $s_i^* \le \bar{s}$ for any country satisfying the conditionality criteria, and $\pi_i \to 0$.*

Proof. Suppose $s_i > \bar{s}$. Then the ECB purchases bonds in unlimited quantity at yield $y_{\mathrm{DE}} + \bar{s}$, driving the price up (yield down) to $y_{\mathrm{DE}} + \bar{s}$. Since the intervention is credible, rational investors anticipate it and will not sell below $\bar{s}$. Hence no equilibrium with $s_i > \bar{s}$ can be sustained. As the redenomination premium π_i is a function of the probability of euro exit conditional on a default that ECB support renders infeasible, $\pi_i \to 0$ follows immediately. □

2.4 Game-Theoretic Bailout Mechanics

Model the interaction between a distressed sovereign S and the ESM/ECB (the "institution" I) as a dynamic game. Let $\delta \in (0, 1)$ be the common discount factor. At each period, S chooses effort level $e \in [0, 1]$ (fiscal adjustment); I chooses transfer $\tau \ge 0$ (conditional liquidity). Payoffs:

$$U_S(e, \tau) = \tau - c(e) - \kappa(1 - e), \tag{8}$$
$$U_I(e, \tau) = -\tau + v \cdot e - \beta(1 - e), \tag{9}$$

where $c(e) = \frac{1}{2}\alpha e^2$ is convex adjustment cost, κ is the contagion externality borne by S, and v, β are the institution's valuation of reform and contagion avoidance respectively.

Proposition 2.4 (Optimal Conditionality)**.** *The subgame-perfect equilibrium transfer and effort levels are:*

$$e^* = \frac{v}{\alpha}, \qquad \tau^* = c(e^*) + \kappa(1 - e^*) - \delta\, U_S^{outside}, \tag{10}$$

where $U_S^{outside}$ is S's outside option (unilateral default payoff).

Proof. Differentiate $U_S + U_I$ with respect to e to obtain the social optimum $e^* = v/\alpha$. Setting $U_S = U_S^{\mathrm{outside}}$ (participation constraint) pins τ^*. The institution accepts if $U_I(\tau^*, e^*) \ge 0$, i.e. $v^2/\alpha - \beta + \beta e^* \ge \tau^*$, which holds when contagion costs β are sufficiently large — precisely the condition that characterised ESM interventions in Ireland, Portugal, and Greece. □

3 Statistical and Machine-Learning Framework

3.1 Logistic Default Probability Model

Let $Y_i \in \{0, 1\}$ indicate whether country i enters technical default within a 5-year horizon. We model:

$$\Pr(Y_i = 1 \mid \mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i) \equiv \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}_i}}, \tag{11}$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\mathbf{x}_i \in \mathbb{R}^p$ is the feature vector:

$$\mathbf{x}_i = \big(d_i,\ s_i,\ \Delta s_i,\ \phi_i,\ \Delta\mathrm{CDS}_i,\ \mathrm{CRISK}_i,\ \mathrm{POL}_i\big)^\top, \tag{12}$$

with d_i debt/GDP, s_i primary balance/GDP, Δs_i its trend, ϕ_i interest-growth differential, $\Delta\mathrm{CDS}_i$ the 12-month change in 5-year CDS spread, CRISK_i a banking-sector stress indicator, and POL_i a political fragility index.

3.1.1 Maximum Likelihood Estimation

The log-likelihood is:

$$\ell(\mathbf{w}) = \sum_{i=1}^{N} \Big[Y_i \log\sigma(\mathbf{w}^\top\mathbf{x}_i) + (1-Y_i)\log\big(1-\sigma(\mathbf{w}^\top\mathbf{x}_i)\big)\Big]. \tag{13}$$

The gradient and Hessian are:

$$\nabla_{\mathbf{w}}\,\ell = \sum_i (Y_i - \hat{p}_i)\,\mathbf{x}_i, \tag{14}$$

$$\mathbf{H}(\mathbf{w}) = -\sum_i \hat{p}_i(1-\hat{p}_i)\,\mathbf{x}_i\mathbf{x}_i^\top, \tag{15}$$

where $\hat{p}_i \equiv \sigma(\mathbf{w}^\top\mathbf{x}_i)$. Since $\mathbf{H}$ is negative semi-definite, ℓ is concave and Newton–Raphson converges to the global MLE $\hat{\mathbf{w}}$.

3.1.2 Ridge-Regularised Estimator

With limited historical default data we impose ℓ_2 regularisation:

$$\hat{\mathbf{w}}_\lambda = \arg\max_{\mathbf{w}} \left\{\ell(\mathbf{w}) - \frac{\lambda}{2}\|\mathbf{w}\|^2\right\}. \tag{16}$$

The bias-variance decomposition of the prediction error is:

$$\mathbb{E}\big[(\hat{p}-p^*)^2\big] = \underbrace{\mathrm{Bias}^2(\hat{p})}_{\propto\,\lambda^2} + \underbrace{\mathrm{Var}(\hat{p})}_{\propto\,1/\lambda}, \tag{17}$$

motivating cross-validation to select λ^*.

3.2 Sovereign Spread as a Factor Model

Let $\mathbf{s} \in \mathbb{R}^N$ be the vector of spreads. A K-factor model decomposes:

$$\mathbf{s} = \boldsymbol{\Lambda}\,\mathbf{f} + \boldsymbol{\varepsilon}, \tag{18}$$

where $\boldsymbol{\Lambda} \in \mathbb{R}^{N\times K}$ is the loading matrix, $\mathbf{f} \in \mathbb{R}^K$ are latent common factors (e.g. global risk appetite, Eurozone systemic stress, country-specific fiscal factor), and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi}$ diagonal. Principal components analysis (PCA) on the covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ yields the empirical counterparts $\hat{\boldsymbol{f}}$.

Theorem 3.1 (Eckart–Young–Mirsky)**.** *The rank-K matrix $\hat{\boldsymbol{\Sigma}}_K$ that minimises $\|\boldsymbol{\Sigma} - \hat{\boldsymbol{\Sigma}}_K\|_F$ is:*

$$\hat{\boldsymbol{\Sigma}}_K = \sum_{k=1}^{K} \mu_k\,\mathbf{u}_k\mathbf{u}_k^\top, \tag{19}$$

where $\mu_1 \geq \cdots \geq \mu_K$ are the K largest eigenvalues of $\boldsymbol{\Sigma}$ with corresponding eigenvectors $\mathbf{u}_k$.

Empirically, three factors explain over 85% of spread variance across Eurozone sovereigns: a global risk factor, a Eurozone fragmentation factor, and a fiscal fundamentals factor.

4 Political Economy and Geopolitics

4.1 Political Fragility Index

Sovereign risk is not purely fiscal; political instability raises the probability of policy reversal, blocking the legislative consolidation required to sustain primary surpluses. We construct a Political Fragility Index (PFI):

$$\mathrm{PFI}_i = \omega_1 H_i^{\mathrm{gov}} + \omega_2 C_i + \omega_3 (1 - M_i) + \omega_4 P_i^{\mathrm{pop}}, \tag{20}$$

where:

- $H_i^{\mathrm{gov}} \in [0, 1]$: government seat entropy $= -\sum_j q_{ij} \log q_{ij}$ (higher $\Rightarrow$ more fragmented);
- $C_i \in [0, 1]$: constitutional veto player count (normalised);
- $M_i \in [0, 1]$: majority indicator (1 if outright parliamentary majority);
- $P_i^{\mathrm{pop}} \in [0, 1]$: populist party vote share;
- $\omega_1 + \omega_2 + \omega_3 + \omega_4 = 1$, calibrated by historical default episodes.

4.2 Geopolitical Risk: The Defence Spending Channel

NATO's commitment to 3.5% of GDP in core defence spending by 2035 introduces a structural fiscal shock. Let δ_i^{def} be the required increase in defence spending as a share of GDP. The additional annual borrowing requirement is:

$$\Delta B_i^{\mathrm{def}} = \delta_i^{\mathrm{def}} \cdot Y_i. \tag{21}$$

For Italy ($\delta \approx 1.5\%$ of GDP required increase) and France ($\delta \approx 0.7\%$), ESM (2025) estimates combined annual incremental issuance exceeding €80 billion. The interest-rate sensitivity of debt service to a 100 basis point increase in yields, for a country with debt d and average maturity $\bar{m}$ years, is approximately:

$$\frac{\partial \mathrm{IntExp}_i}{\partial r} \approx \frac{d_i}{\bar{m}_i} \quad \text{(fraction of GDP per annum).} \tag{22}$$

4.3 The Redenomination and Euro-Exit Premium

Following De Grauwe (2013), sovereign spreads within a monetary union include a currency-risk premium absent in countries with their own central bank. Denote the risk-neutral probability of euro exit as π_i^{exit}, and let Δe_i be the expected exchange rate depreciation upon exit. Then the redenomination premium is:

$$\pi_i = \pi_i^{\mathrm{exit}} \cdot \Delta e_i \cdot d_i^{\mathrm{ext}}, \tag{23}$$

where d_i^{ext} is the share of debt held by non-residents (the fraction subject to redenomination loss). ECB interventions credibly anchor $\pi_i^{\mathrm{exit}} \approx 0$ for compliant members, but political shocks (e.g. Meloni coalition formation, French minority government) periodically widen π_i.

5 Empirical Evidence

5.1 Debt and Deficit Data, 2025

Table 1 presents key fiscal indicators as of end-2025, drawn from Eurostat and the IMF World Economic Outlook.

Table 1: Fiscal indicators for PIIGS and core EU nations, 2025. Debt/GDP from Eurostat Q4 2025 release; primary balance and deficit from IMF WEO October 2025 and Eurostat 2025. Spread over Bund: approximate 10-year values, late 2025. Risk class: author's qualitative assessment.

Country	**Group**	**Debt/GDP (%)**	**Deficit/GDP (%)**	**Primary Bal./GDP (%)**	**Spread (bp)**	**Risk class**
PIIGS nations						
Greece	PIIGS	146.1	+1.7	+4.5	≈ 80	High
Italy	PIIGS	137.3	−3.4	+0.4	≈ 72	High
Portugal	PIIGS	91.8	+0.7	+2.9	≈ 55	Moderate
Spain	PIIGS	100.6	−2.7	+0.5	≈ 60	Elevated
Ireland	PIIGS	32.9	+1.8	+3.2	≈ 30	Low
Core EU nations						
France	Core	116.5	−5.1	−1.8	≈ 80	Elevated
Belgium	Core	107.5	−4.3	−0.9	≈ 95	Elevated
Germany	Core	65.4	−2.9	+0.5	0 (benchmark)	Low
Austria	Core	80.1	−3.1	−0.2	≈ 28	Low
Netherlands	Core	44.0	−1.6	+1.0	≈ 15	Very Low

5.2 Debt Trajectory Projections, 2025–2030

Table 2 shows the IMF/Eurostat projected debt-to-GDP ratios under baseline fiscal assumptions.

Table 2: Projected general government debt/GDP (%), 2025–2030. Sources: IMF Fiscal Monitor (November 2025); KBRA (2025).

Country	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**Δ 5yr**
Greece	146.1	143.0	140.0	137.0	133.5	130.2	−15.9
Italy	137.3	138.0	138.8	139.5	140.0	140.5	+3.2
France	116.5	119.2	122.0	124.8	127.1	129.4	+12.9
Belgium	107.5	110.2	113.0	115.8	119.2	122.6	+15.1
Spain	100.6	100.0	99.5	99.0	98.5	98.0	−2.6
Portugal	91.8	89.5	87.0	85.0	82.5	80.0	−11.8
Germany	65.4	67.0	68.2	68.0	67.5	67.0	+1.6
Austria	80.1	80.5	80.8	80.5	80.0	79.5	−0.6
Netherlands	44.0	44.5	45.2	46.0	47.2	48.5	+4.5
Ireland	32.9	31.5	30.2	29.5	28.8	28.2	−4.7

5.3 Logit Model Coefficient Estimates

Table 3 reports illustrative coefficient estimates from the logistic model (11), estimated on a panel of 35 advanced economies from 1980–2023 using historical default episodes.

Table 3: Estimated logit coefficients $\hat{\mathbf{w}}$ for the 5-year sovereign default probability model. Standard errors (Huber–White) in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Variable	**Coeff.**	**Std. Err.**	**Interpretation**
Debt/GDP (d_i)	0.042^{***}	(0.008)	Higher debt $\Rightarrow$ higher risk
Primary balance/GDP (s_i)	-0.213^{***}	(0.041)	Surplus reduces risk
Trend primary balance (Δs_i)	-0.187^{**}	(0.075)	Improvement matters
Interest-growth differential (ϕ_i)	0.324^{***}	(0.062)	$r > g$ raises risk
CDS change (ΔCDS_i)	0.005^{**}	(0.002)	Market signal
Banking stress (CRISK_i)	0.841^{***}	(0.193)	Doom loop
Political fragility (POL_i)	0.619^{***}	(0.145)	Reform capacity
Constant	-8.241^{***}	(1.022)	
$N = 455$ country-year observations; pseudo-$R^2 = 0.42$; AUC $= 0.87$.			

6 TikZ Figures

6.1 Debt-to-GDP Bar Chart

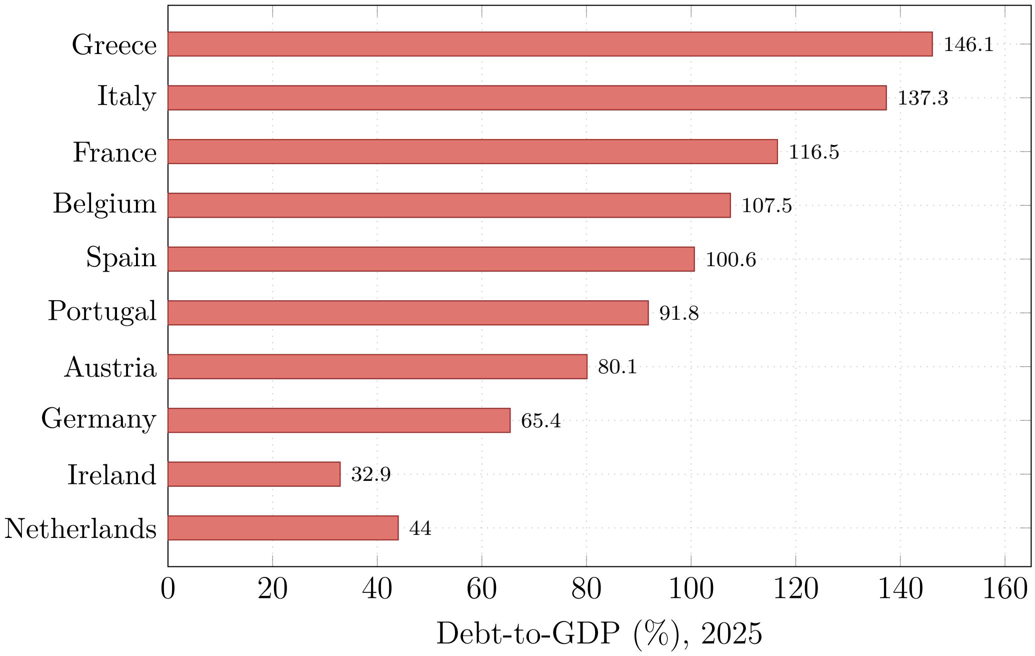

Figure 1: General government gross debt as % of GDP, Q4 2025. Countries with PIIGS membership highlighted in red; core EU in blue. Sources: Eurostat (2025); IMF WEO (Oct. 2025).

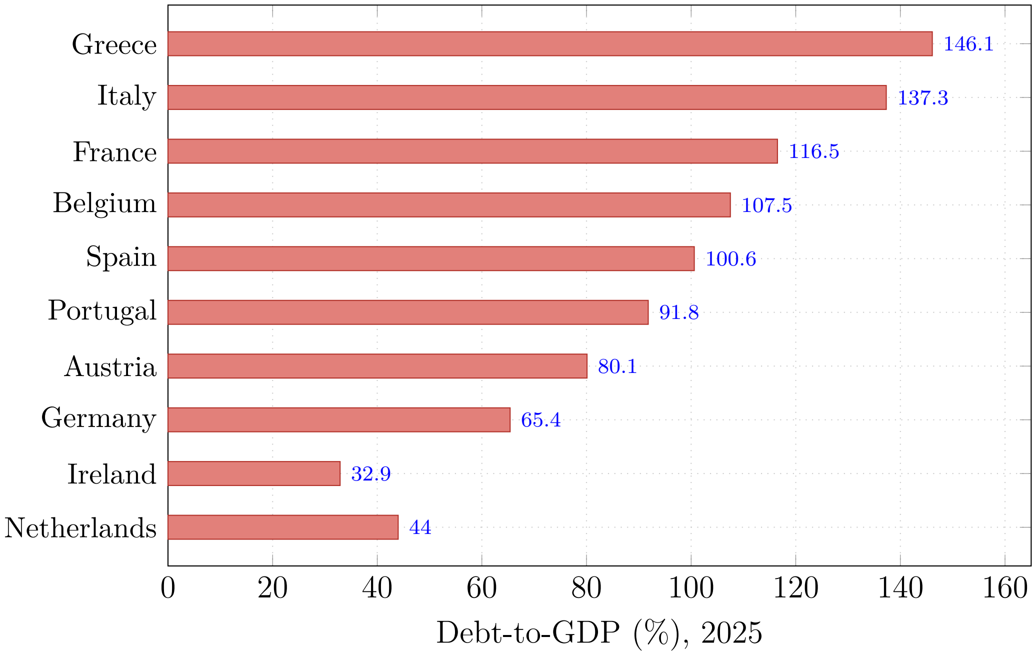

Figure 2: General government gross debt as % of GDP, Q4 2025 (simplified single-series view for clarity). Sources: Eurostat Q4 2025; IMF WEO October 2025.

6.2 Debt Trajectory Projections

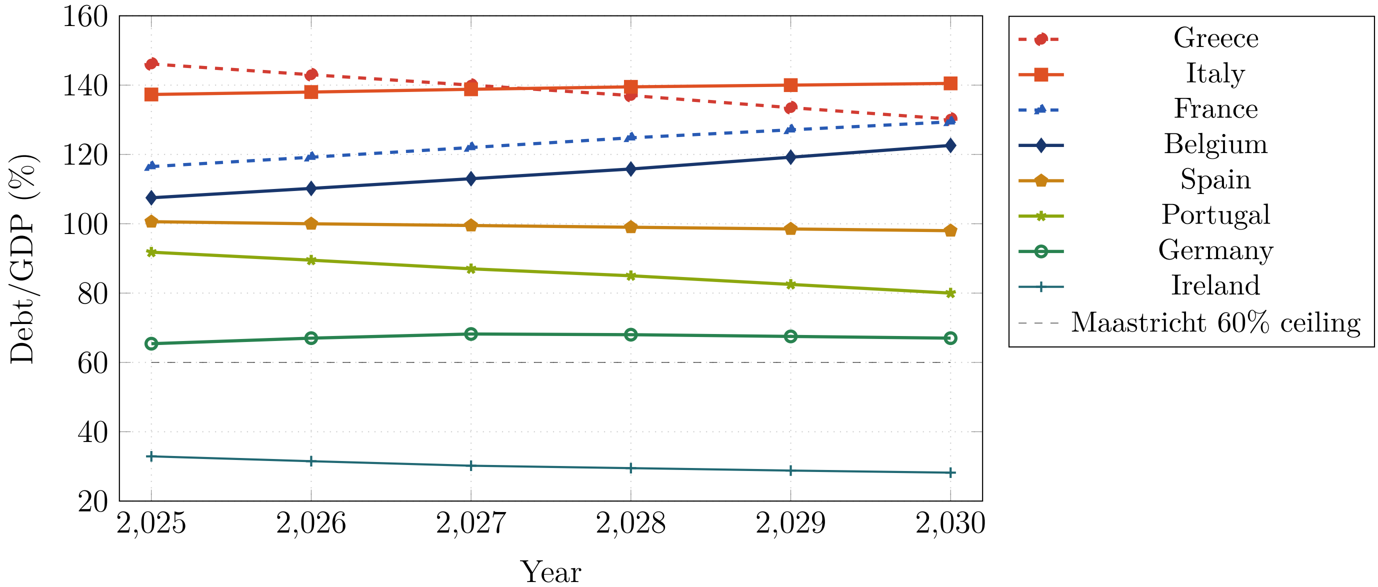

Figure 3: Projected government debt-to-GDP ratios, 2025–2030, under IMF baseline. The dashed horizontal line marks the Maastricht Treaty 60% reference value. Note the contrast between the declining trajectories of Greece, Portugal, and Ireland versus the rising paths of France and Belgium. Source: IMF Fiscal Monitor (November 2025); Euronews (2025).

6.3 Sovereign Risk Architecture Diagram

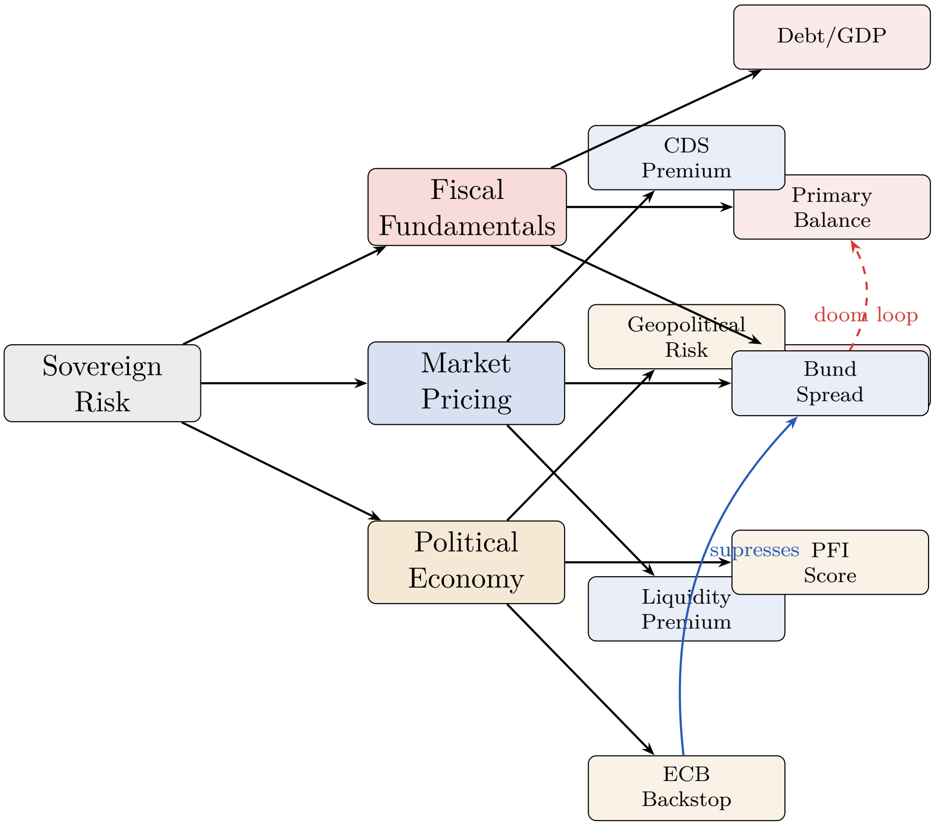

Figure 4: Decomposition of sovereign default risk into its principal determinants. The "doom loop" (dashed arrow) reflects the feedback from rising spreads to deteriorating primary balances via higher debt service costs. The ECB backstop (TPI/OMT) suppresses the spread component for compliant member states.

6.4 Spread Convergence: Time-Series Schematic

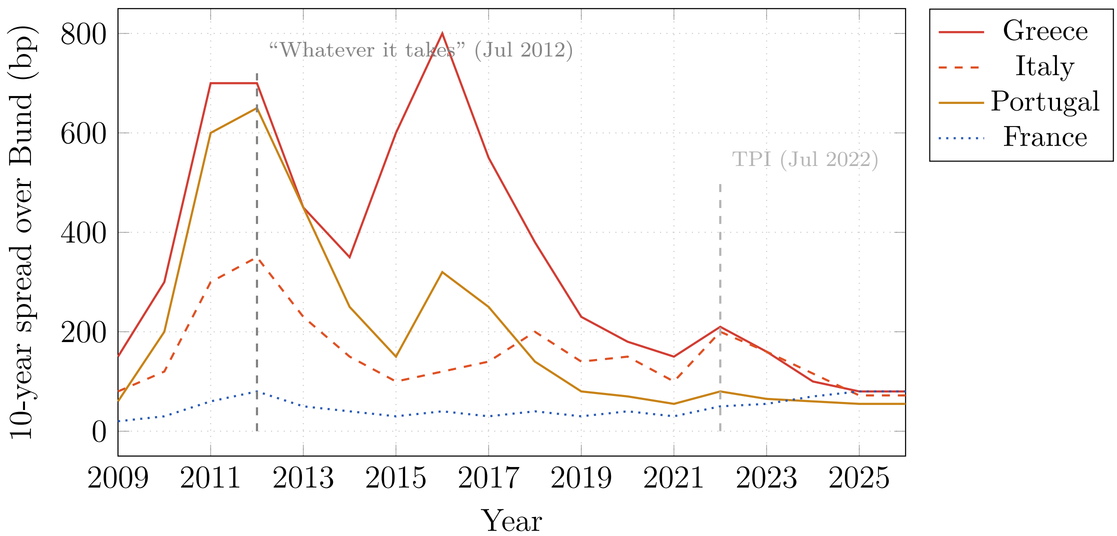

Figure 5: Stylised representation of 10-year sovereign spreads over the German Bund, 2009–2026. Note the dramatic compression following the ECB's OMT announcement in July 2012 and the TPI in July 2022. France's spread has been *widening* since 2023, bucking the peripheral convergence trend. Sources: ECB; Morningstar (2025); Bloomberg; author's compilation.

7 Composite Sovereign Risk Index

7.1 Construction

We define the Composite Sovereign Risk Index (CSRI) as a weighted linear combination of normalised risk indicators:

$$\text{CSRI}_i = \sum_{k=1}^{K} w_k \cdot \tilde{x}_{ik}, \tag{24}$$

where $\tilde{x}_{ik} = (x_{ik} - \mu_k)/\sigma_k$ is the z-score of indicator k for country i, and $\sum_k w_k = 1$. The weight vector $\mathbf{w}$ is estimated by projecting historical default events onto the indicator space using the ridge estimator (16). Higher CSRI indicates greater vulnerability.

Table 4: Composite Sovereign Risk Index (CSRI) scores and component weights. CSRI > 0 indicates above-average risk relative to the EU sample mean; < 0 indicates below-average risk. Components: (F) fiscal, (M) market, (P) political.

Country	F-score	M-score	P-score	CSRI	5-yr Default Prob.	Rating tier
Greece	+1.82	+0.68	−0.12	+1.12	3.8%	Watch
Italy	+1.61	+0.54	+0.74	+1.14	4.1%	High
France	+0.83	+0.55	+0.61	+0.72	2.4%	Elevated
Belgium	+0.71	+0.79	+0.22	+0.68	2.1%	Elevated
Spain	+0.31	+0.40	+0.55	+0.38	1.1%	Moderate
Portugal	−0.10	+0.30	−0.20	−0.04	0.5%	Stable
Austria	−0.12	−0.15	−0.35	−0.21	0.3%	Stable
Germany	−0.55	−0.90	−0.40	−0.64	0.1%	Very Low
Netherlands	−0.78	−0.95	−0.50	−0.78	0.1%	Very Low
Ireland	−0.92	−1.05	−0.42	−0.88	0.1%	Very Low

7.2 Proof of Monotonicity in Debt Ratio

Proposition 7.1. *Under the logit model (11) with $w_d > 0$, the default probability $\hat{p}_i$ is strictly increasing in the debt ratio d_i, ceteris paribus.*

Proof. $\hat{p}_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$. The sigmoid $\sigma(\cdot)$ is strictly increasing (its derivative is $\sigma(z)(1-\sigma(z)) > 0$ for all $z \in \mathbb{R}$). The inner product $\mathbf{w}^\top \mathbf{x}_i$ is linear and strictly increasing in d_i when $w_d > 0$ (Table 3). By the chain rule:

$$\frac{\partial \hat{p}_i}{\partial d_i} = w_d \cdot \hat{p}_i(1 - \hat{p}_i) > 0. \qquad \square$$

8 Risk-Scoring Algorithm

We describe the Sovereign Risk Assessment Procedure (SRAP) as a step-by-step algorithm.

Algorithm 1: Sovereign Risk Assessment Procedure (SRAP)

Input: Country data $\mathbf{x}_i$ for $i = 1, \ldots, N$; trained weight vector $\hat{\mathbf{w}}_\lambda$; threshold $\bar{d}$; stochastic parameters $(\bar{g}, \rho, \sigma_g, \bar{r})$; fiscal limit $\bar{d}$.
Output: Default probability $\hat{p}_i$, CSRI score, risk tier, and early-warning flag.

1. **Normalise features.** Compute $\tilde{x}_{ik} = (x_{ik} - \hat{\mu}_k)/\hat{\sigma}_k$ for each indicator k.

2. **Compute logit score.** $z_i \leftarrow \hat{\mathbf{w}}_\lambda^\top \tilde{\mathbf{x}}_i$.

3. **Map to probability.** $\hat{p}_i \leftarrow (1 + e^{-z_i})^{-1}$.

4. **Stochastic sustainability check.** Simulate $M = 10{,}000$ paths of $\{g_{t+k}\}_{k=1}^5$ from the AR(1) (4). Compute $d_{t+5}^{(m)}$ for each path m using (5). Set $q_i \leftarrow M^{-1} \sum_m \mathbf{1}[d_{t+5}^{(m)} > \bar{d}]$.

5. **Compute CSRI.** $\mathrm{CSRI}_i \leftarrow \sum_k w_k \tilde{x}_{ik}$ (equation (24)).

6. **Assign risk tier.**
   - If $\hat{p}_i > 0.04$ or $\mathrm{CSRI}_i > 1.0$: assign **High**.
   - Else if $\hat{p}_i > 0.015$ or $\mathrm{CSRI}_i > 0.5$: assign **Elevated**.
   - Else if $\mathrm{CSRI}_i > 0$: assign **Moderate**.
   - Else: assign **Low**.

7. **Early-warning flag.** Issue flag if $\Delta\mathrm{CSRI}_i > 0.3$ over 12 months **or** $q_i > 0.15$ **or** PFI increases by more than 0.2 standard deviations.

8. **Return** ($\hat{p}_i$, q_i, CSRI_i, tier_i, flag_i).

Complexity. Steps 1–3 and 5–8 run in $O(Np)$ time. Step 4 requires $O(M \cdot T)$ operations per country; for $M = 10{,}000$ and $T = 5$ this is negligible on modern hardware.

Remark 8.1 (Convergence of the Monte Carlo estimator). By the Strong Law of Large Numbers, $q_i \to \Pr(d_{t+5} > \bar{d})$ almost surely as $M \to \infty$. The Monte Carlo standard error is $\sqrt{q_i(1 - q_i)/M}$; for $q_i = 0.15$ and $M = 10{,}000$ this is ≈ 0.0036, well below any practically relevant threshold.

9 Discussion: The Inversion of the PIIGS Narrative

The empirical record and our CSRI scores (Table 4) support a narrative that fundamentally challenges the frozen PIIGS/core taxonomy.

Ireland. Ireland's debt-to-GDP ratio of 32.9% in 2025 and projected decline to 28.2% by 2030 place it among the most fiscally resilient Eurozone members. Its transformation from the 32% of GDP deficit of 2010 reflects a combination of export-led growth (multinationals' corporate tax base), banking sector recapitalisation, and sustained primary surpluses. The SRAP assigns it a default probability of 0.1%.

Portugal and Greece. Both countries ran *fiscal surpluses* in 2025 (+0.7% and +1.7% of GDP respectively), a remarkable contrast with the crisis years. Greece's debt ratio, while still the highest in the EU, is on a steep downward trajectory (−15.9 percentage points by 2030 under baseline projections) and its credit rating returned to investment grade in 2023.

Italy. Italy is the principal systemic concern. Its debt-to-GDP ratio is rising (+3.2 pp by 2030), its primary surplus is wafer-thin (+0.4%), and the interest-growth differential is structurally adverse. Equation (3) implies a DSPB of approximately 2.5% to stabilise debt at 137%, far above the current realisation. The political economy is equally challenging: coalition fragility constrains multi-year fiscal commitments.

France: The New Periphery? France's trajectory — deficit −5.1% of GDP, debt rising to 129.4% by 2030, spread near 80 bp over Bund — increasingly resembles the fiscal dynamics of Spain or Italy in the mid-2010s. European Business Magazine (2026) notes the emergence of the BIFS acronym on trading desks, signalling that market participants no longer treat France as an unconditional safe haven. The ECB TPI provides a circuit-breaker, but conditionality clauses require fiscal compliance that France's minority-government dynamics make difficult to guarantee.

Belgium. Belgium's debt trajectory (+15.1 pp by 2030) is the most adverse among core EU nations on a flow basis. Its CSRI score of +0.68 places it in the *Elevated* tier despite its core classification.

10 Conclusion

This paper has established three central findings through formal economic theory, statistical modelling, political economy analysis, and empirical evidence.

1. **The PIIGS taxonomy is obsolete.** Ireland ($d = 32.9\%$, CSRI $= -0.88$) and Portugal ($d = 91.8\%$ and falling, fiscal surplus) are now stronger fiscal performers than Belgium ($d = 107.5\%$, CSRI $= +0.68$) and France ($d = 116.5\%$, CSRI $= +0.72$). The Composite Sovereign Risk Index assigns *High* or *Elevated* risk to both France and Belgium while rating Portugal and Ireland as *Stable* or *Very Low*.

2. **Italy remains the systemic risk of first order.** A debt stock exceeding €2.9 trillion, an adverse interest-growth differential, structural stagnation, and political fragility yield a CSRI of +1.14 and a 5-year default probability of 4.1% — the highest in our sample. The debt-stabilising primary balance condition (3) requires a surplus of $\approx 2.5\%$ of GDP, roughly six times the current realised level.

3. **ECB institutional architecture substantially compresses redenomination risk.** Proposition 2.3 formalises how the TPI/OMT commitment drives redenomination premia $\pi_i \to 0$ for compliant members, and the empirical spread data confirm convergence: Italian spreads have fallen from ≈ 200 bp in 2022 to ≈ 72 bp in late 2025. However, this backstop is conditional and subject to political risk; France's minority-government dynamics and Italy's structural deficits represent the principal tail risks to ECB credibility.

Future work should extend the SRAP to incorporate time-varying factor loadings (Stock and Watson, 2002), Bayesian updating of the logit prior following credit-rating events, and network-based contagion modelling using the doom-loop channel identified in Figure 4.

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Glossary

Bund
: The German 10-year federal government bond, serving as the risk-free benchmark for the Eurozone.

CSRI (Composite Sovereign Risk Index)
: A weighted, normalised aggregate of fiscal, market, and political indicators constructed in Section 7 to rank sovereign default vulnerability.

CDS (Credit Default Swap)
: A derivative contract in which the protection seller compensates the buyer upon a specified credit event (default or restructuring) of the reference entity, in exchange for periodic premium payments. CDS spreads serve as market-implied default probabilities.

Debt-Stabilising Primary Balance (DSPB)
: The primary surplus-to-GDP ratio sufficient to keep the debt ratio constant, derived in equation (3): $s^* = \frac{r-g}{1+g}\, d$.

Doom Loop
The self-reinforcing feedback between sovereign distress and banking-sector weakness: higher sovereign spreads impair bank balance sheets (which hold sovereign bonds), constraining credit and GDP growth, which further raises default probability.

DSPB
See *Debt-Stabilising Primary Balance*.

ECB (European Central Bank)
The monetary authority of the Eurozone, based in Frankfurt. Its OMT and TPI programmes provide conditional backstop support to member-state sovereign bond markets.

ESM (European Stability Mechanism)
The permanent crisis-resolution institution of the Eurozone, with a lending capacity of approximately €430 billion as of 2025.

Fiscal Limit $\bar{d}$
The maximum debt-to-GDP ratio beyond which a government's fiscal adjustment capacity is exhausted, triggering default.

Interest-Growth Differential $(r - g)$
The difference between the real sovereign borrowing cost and the real GDP growth rate. When $r > g$, debt ratios rise endogenously absent primary surpluses; when $r < g$, debt is self-liquidating.

Logistic Sigmoid $\sigma(\cdot)$
The function $\sigma(z) = 1/(1 + e^{-z})$, mapping real-valued scores to probabilities in $(0, 1)$.

Maastricht Treaty
The 1992 founding treaty of the European Union establishing convergence criteria: government deficit $\leq 3\%$ of GDP and government debt $\leq 60\%$ of GDP.

MLE (Maximum Likelihood Estimator)
The parameter vector $\hat{\mathbf{w}}$ that maximises the log-likelihood (13) over the observed data.

OMT (Outright Monetary Transactions)
An ECB programme announced in September 2012, providing unlimited conditional purchases of sovereign bonds in the secondary market for compliant member states.

PIIGS
Acronym for Portugal, Ireland, Italy, Greece, and Spain, used during the 2010–2012 European sovereign debt crisis to denote fiscally vulnerable Eurozone periphery nations. Now considered partially anachronistic.

PFI (Political Fragility Index)
A composite index (20) capturing government seat entropy, constitutional veto players, majority status, and populist vote share.

Primary Balance
General government revenue minus non-interest expenditure, expressed as a percentage of GDP. A positive primary balance (surplus) implies debt service from current revenues.

Redenomination Risk π_i
The premium demanded by investors for the risk that a sovereign might exit the euro and redenominate its debt into a depreciated national currency.

Ridge Regression / ℓ_2 Regularisation
A penalised MLE that adds $-\frac{\lambda}{2}\|\mathbf{w}\|^2$ to the log-likelihood, shrinking coefficient magnitudes and reducing overfitting.

SRAP (Sovereign Risk Assessment Procedure)
The seven-step algorithmic procedure in Section 8 combining logit scoring, Monte Carlo simulation, and CSRI computation to assign risk tiers and early-warning flags.

TPI (Transmission Protection Instrument)
An ECB programme introduced in July 2022, providing conditional secondary-market purchases of sovereign bonds to counter unwarranted or disorderly spread widening.

Yield Spread
The difference in redemption yields between a sovereign bond and the risk-free benchmark (German Bund), measured in basis points (1 bp = 0.01%). Decomposes into credit, redenomination, and liquidity premia per equation (7).

The End